

COLE SHANKS

Electrical engineering graduate from UBC, focused on embedded systems and digital hardware — firmware in C, FPGA and Verilog, and hands-on hardware bring-up and validation, including a capstone building the control firmware for an autonomous field robot. Based in Japan, native English speaker with functional Japanese (JLPT N3 / CEFR B1), looking for embedded or software roles, with a current focus on Rust.

cshanks223@gmail.com / coleshanks.github.io / Kure, Hiroshima, Japan

EDUCATION

BASc Electrical Engineering, 2018–2022

University of British Columbia

Graduated with distinction · Minor in Commerce

Main degree — digital systems design, microcomputers, signals & systems, control systems, electronics, and a year-long capstone.

BASc Engineering Transfer Program, 2017–2018

Capilano University

First-year engineering transfer — calculus, physics, and chemistry foundations before transferring to UBC.

BCom Accounting and Finance, 2016–2017

University of Toronto

First year of Rotman Commerce before switching into electrical engineering.

EXPERIENCE

Assistant Language Teacher

JET Programme

Aug 2022–Present, Kure, Hiroshima, Japan

- Plan and deliver English lessons, teaching and presenting in front of classes.
- Coordinate daily with Japanese teaching staff; coach students for English speech contests.
- Take part in school life — sports day, culture festival, and other events.
- Developed working Japanese (JLPT N3 / CEFR B1) in a full-time Japanese-language office environment.

Application Engineer (Co-op)

Maxim Integrated

Jun 2020–Dec 2020, Burnaby, BC, Canada

- Reliability and soak testing of USB/DisplayPort extension hardware — power-cycle, hot-plug, and multi-day endurance runs with pass/fail and error logging.
- Ran a peripheral compatibility matrix (conferencing cameras, speakerphones, mics, storage) across multiple host setups.
- Validated links in live video calls; measured throughput and data integrity with iperf3 and PassMark BurnInTest.

PROJECTS

monoiten — command-line monolingual Japanese dictionary, built in Rust (2026, ongoing)

Propbot — Autonomous Data-Collection Robot — ECE capstone: a self-driving robot that collects wireless-signal data in the field (2022)

Simple RISC Machine — a CPU designed from scratch in Verilog and run on an FPGA (2019)

ELEC 291 — Coin-Picking Robot — a battery-powered robot that finds and collects coins while staying inside a marked perimeter (2019)

ELEC 291 — Reflow Oven Controller — 8051-assembly firmware that runs a solder-reflow oven through a set temperature profile, with an LCD and button interface (2019)

SKILLS

Rust · Low-level (C, 8051 assembly) · OOP (C++, C#) · HDL / FPGA (Verilog) · Web (HTML, CSS)
git / GitHub · Unix / CLI (fish, WSL) · VS Code / Vim / Visual Studio · PlatformIO
UART · Quartus / ModelSim · MATLAB / Simulink · Oscilloscope / Function Generator

LANGUAGES

English — native

Japanese — JLPT N3 · CEFR B1 2026